Homework Section 15.6

1. Evaluate the iterated integrals:

   a) $\int_{-1}^{1}\int_{0}^{2}\int_{0}^{1}(xz-y^3)\,dzdydx$ b) $\int_{0}^{3}\int_{0}^{1}\int_{0}^{\sqrt{1-z^2}} ze^y\,dxdzdy$

2. Evaluate the triple integrals:

   a) $\iiint\limits_Q yz\cos(x^5)\,dV$, where $Q=\{(x,y,z)\,|\,0\le x\le 1,\ 0\le y\le x,\ x\le z\le 2x\}$

   b) $\iiint\limits_Q xy\,dV$, where Q is the solid tetrahedron with vertices (0 ,0, 0), (1, 0, 0), (0, 2, 0), and (0, 0, 3).

   c) $\iiint\limits_Q (x+2y)\,dV$, where Q is bounded by the parabolic cylinder $y=x^2$ and the planes $x=z$, $x=y$, and $z=0$.

   d) $\iiint\limits_Q x\,dV$, where Q is bounded by the paraboloid $x=4y^2+4z^2$ and the plane $x=4$.

3. Use a triple integral to find the volume of the given solid:

   a) The solid bounded by the cylinder $y=x^2$ and the planes $z=0$, $z=4$, and $y=9$.

   b) The solid enclosed by the cylinder $x^2+y^2=9$ and the planes $y+z=5$ and $z=1$.

4. Sketch the solid whose volume is given by the iterated integral:

   a) $\int_{0}^{1}\int_{0}^{1-x}\int_{0}^{2-2z} dydzdx$ b) $\int_{0}^{2}\int_{0}^{2-y}\int_{0}^{4-y^2} dxdzdy$

5. Express the integral $\iiint\limits_Q f(x,y,z)\,dV$ as an iterated integral in six different ways, where Q is the solid bounded by the surfaces $z=0$, $z=y$, $x^2=1-y$.

6. Consider the iterated integral $\int_{0}^{1}\int_{\sqrt{x}}^{1}\int_{0}^{1-y} f(x,y,z)\,dzdydx$. Rewrite this integral as an equivalent integral in the five other orders.